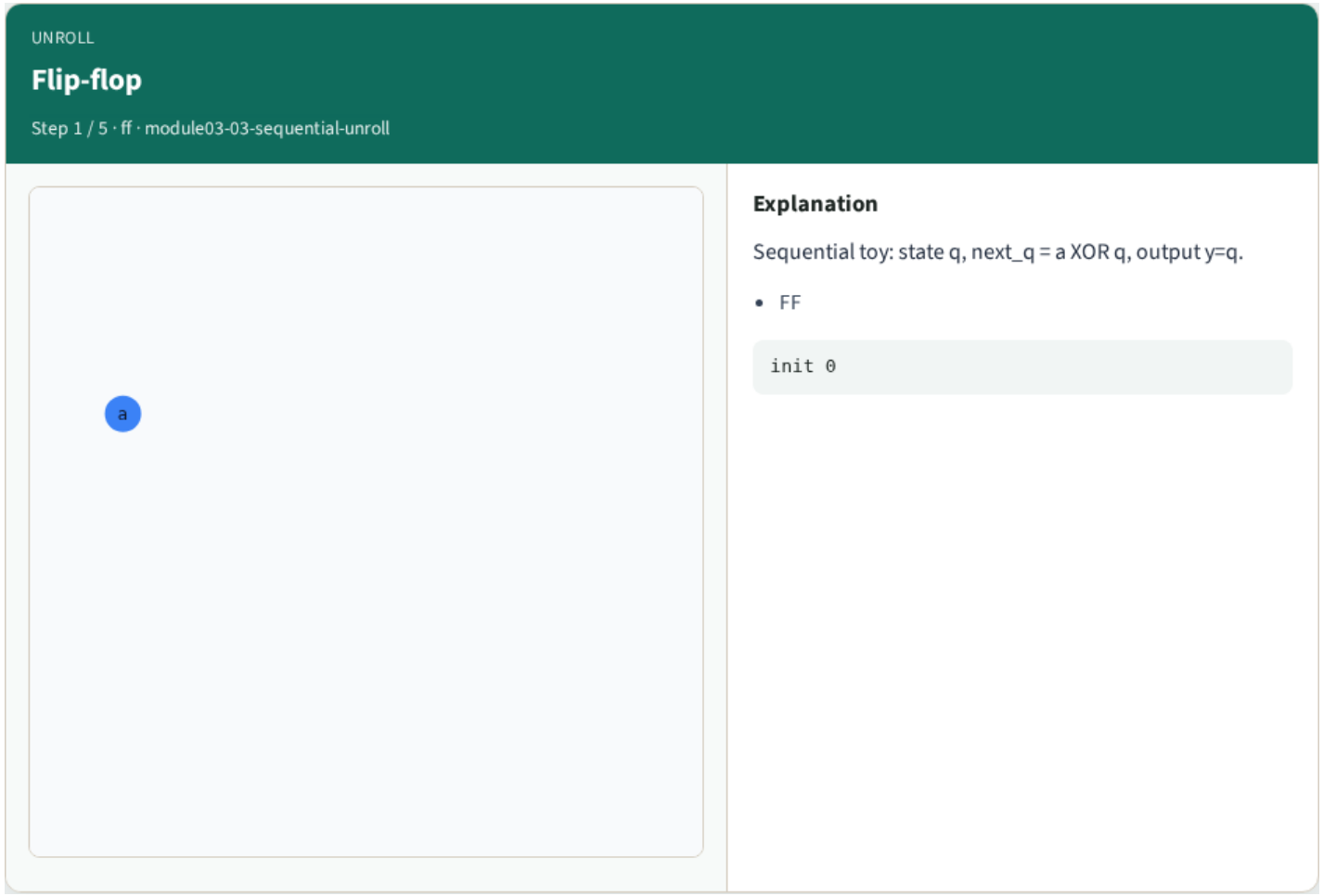
Seq unroll

Sequential equivalence often unrolls both designs for k cycles and runs combinational CEC on the unrolled POs

The idea

- Sequential_unroll(spec

Flip-flop



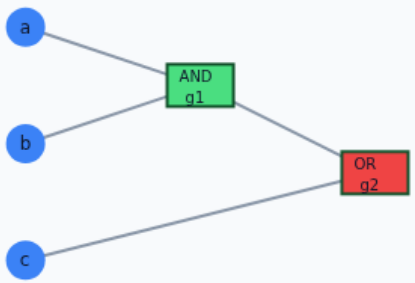
Flip-flop

Unroll k

UNROLL

Unroll k

Step 2 / 5 · steps · module03-03-sequential-unroll



Explanation

Unrolling k steps builds a combo circuit over time frames.

- frames

k=2

Unroll k

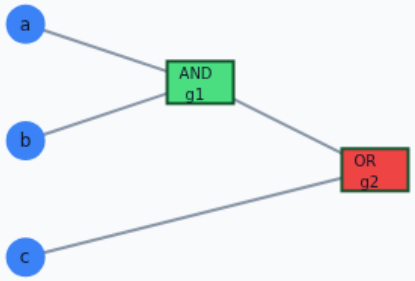
Frame PIs

UNROLL

Frame PIs

Step 3 / 5 · pis · module03-03-sequential-unroll

unroll pis=3



Explanation

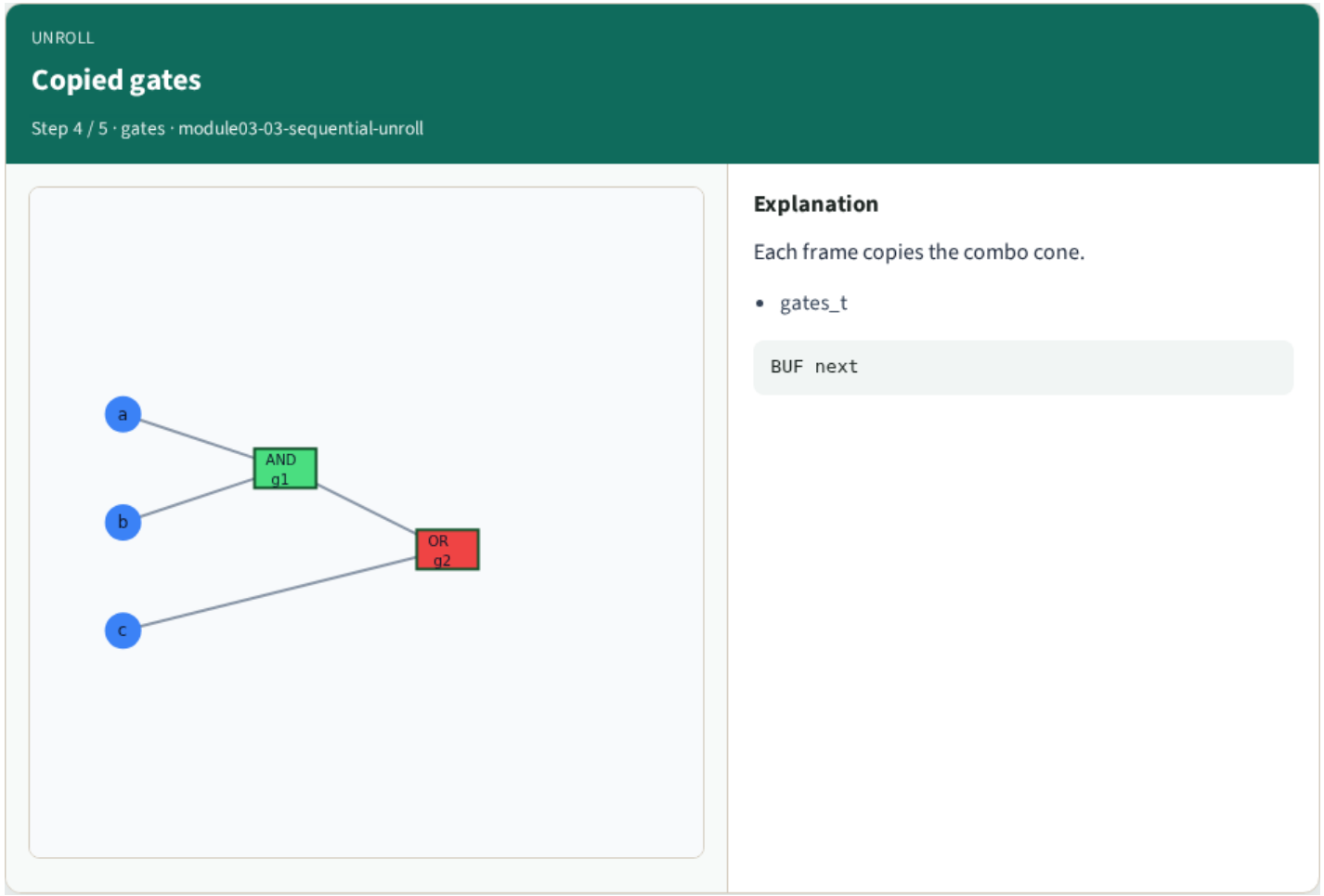
Unroll adds a_0, a_1, … and q_0 as inputs.

- time

```
frames
```

Frame PIs

Copied gates



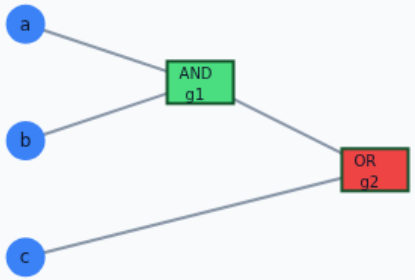
Copied gates

Why unroll

UNROLL

Why unroll

Step 5 / 5 · why · module03-03-sequential-unroll



Explanation

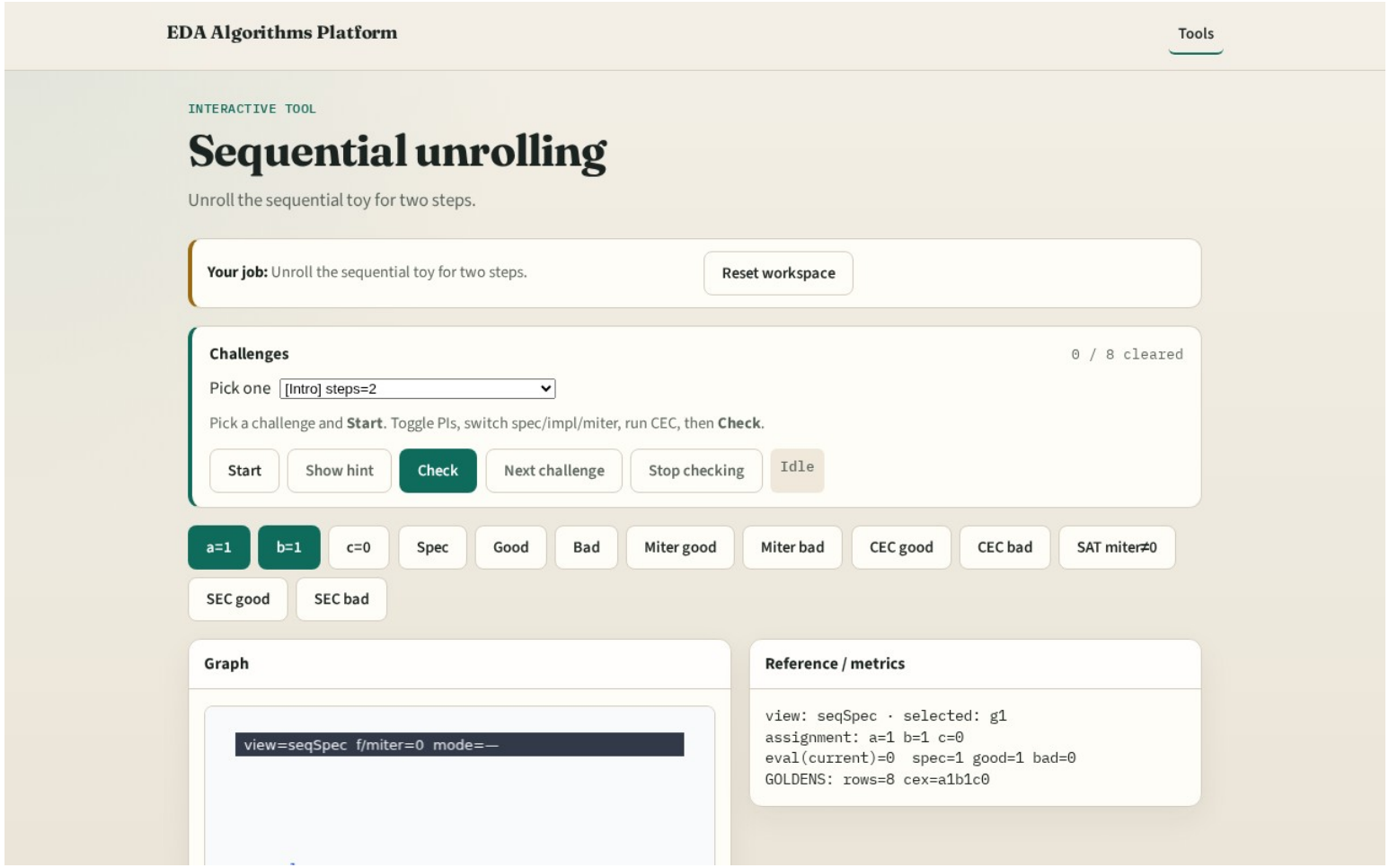
Bounded SEC compares unrolled behaviors for k cycles.

- BMC-ish

```
Next SEC
```

Why unroll

Browser lab track



Browser lab starter

Implement track

- Call sequential_unroll(DATA['sequential']['spec'], 2); inspect gates and PIs

Pitfalls

- Unrolling without resetting state
- Confusing PO y with next state

Your turn

- Next: sequential CEC lite